Traffic noise and housing

About one in three Danish homes is exposed to road noise above the recommended limit. Traffic noise disturbs sleep, increases the risk of cardiovascular disease and costs society billions. The course “Traffic noise and housing” was about calculating the noise from roads and railways and protecting homes from it. The page has an interactive road with cars and a noise barrier.

> DTH, course 5148 (1988), 3 points: Traffic noise and housing, Acoustics Laboratory.

The source: tyres and engine

A car’s noise has two main sources. At low speed and when accelerating, engine, intake and exhaust dominate. Above about 30–40 km/h for cars, *tyre/road noise* dominates: air squeezed out of and sucked into the tread pattern (“air pumping”), and vibrations in the tyre as the tread blocks hit the road. Tyre/road noise grows with speed roughly as $30 \log_{10} v$ – doubling the speed gives 9 dB more. That is why electric cars help only a little on motorways, but quite a lot in town.

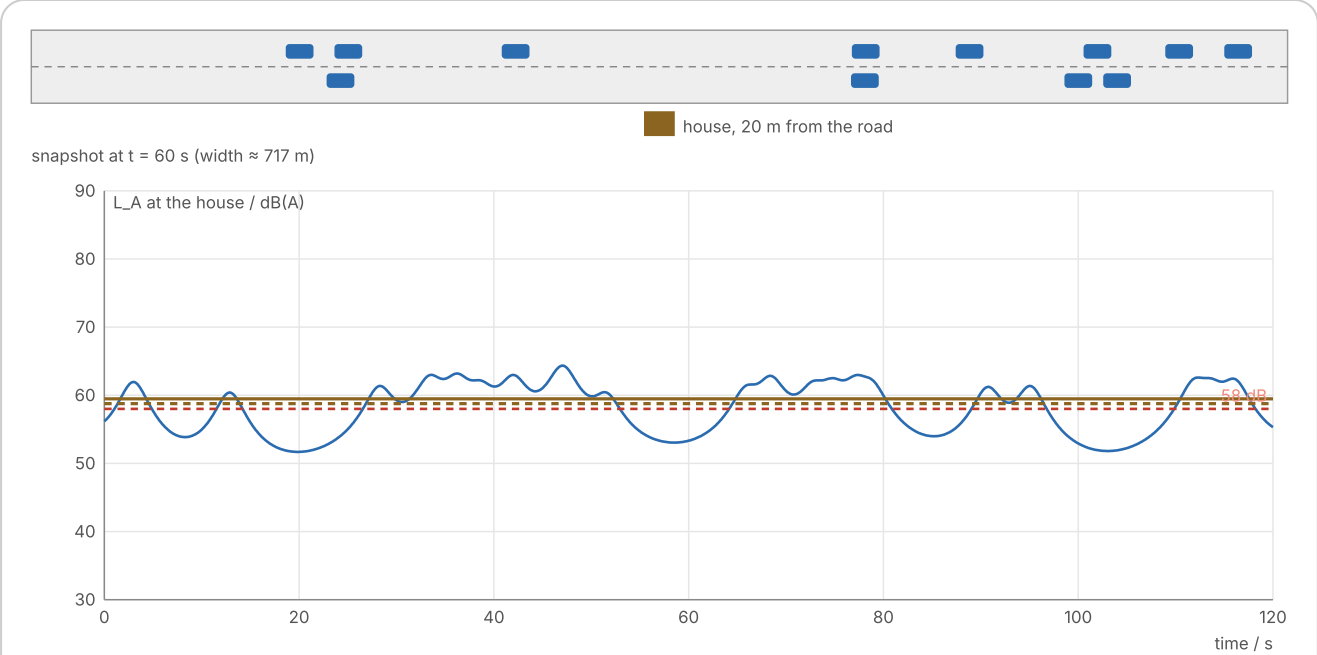
Heavy vehicles make as much noise as 5–10 cars. A single lorry at night can wake more people than a hundred cars by day – which is why night-time noise is weighted specially.

From car to road: the line source

One passing car is a moving point source. The sound pressure rises as it approaches and falls once it has passed. A road with many cars behaves over time like a *line source*: the sound power per metre of road is constant, and it spreads over a cylinder, not a sphere. The equivalent level therefore falls by only 3 dB per doubling of distance, not 6 dB. Integrating over the passage of a car with sound power W on a hard surface gives

$$L_{\mathrm{eq}} = L_W + 10 \log_{10} \frac{N}{2\, d\, v \cdot 3600\ \mathrm{s}}$$

where N is the number of cars per hour, d the distance and v the speed. Twice as many cars give 3 dB more. Half the speed gives fewer decibels from each car, but the cars are present for twice as long, so the net effect with the $30 \log v$ law is about 6 dB.



Indicators and limits

Traffic noise is stated as a day-weighted equivalent level, L_{den} (day–evening–night), where evening noise (19–22 h) gets a penalty of 5 dB and night noise (22–07 h) 10 dB:

$$L_{\mathrm{den}} = 10\log_{10}\frac{1}{24}\left(12\cdot 10^{L_d/10} + 3\cdot 10^{(L_e+5)/10} + 9\cdot 10^{(L_n+10)/10}\right)$$

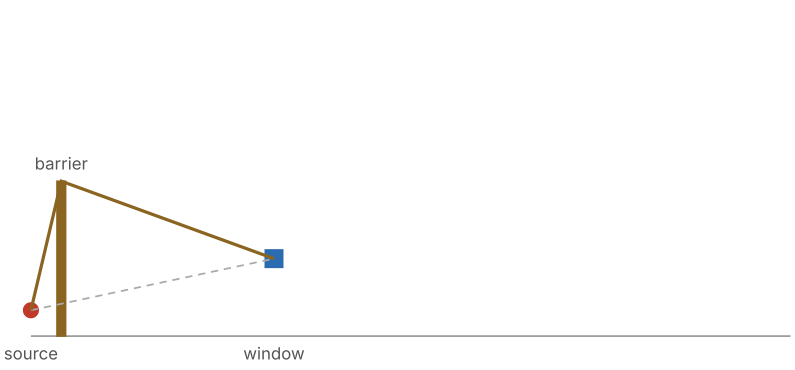
The Danish Environmental Protection Agency's guideline limit for road noise at homes is $L_{\mathrm{den}} = 58$ dB outdoors and 33 dB indoors with the windows closed. In the 1980s $L_{\mathrm{Aeq,24h}}$ was used, with a limit of 55 dB. Calculations follow common Nordic models: the Nordic road-traffic model of 1996 and the more detailed Nord2000, which takes account of the type of ground, wind and temperature gradients in the atmosphere.

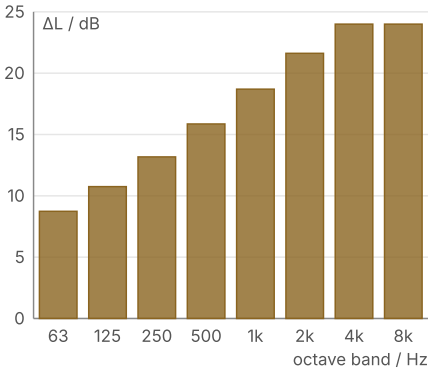
Noise barriers

A barrier works by forcing the sound to go over the top. Behind the barrier there is a shadow, but the sound bends into it at the edge – diffraction, exactly like light at an edge. How much the barrier attenuates depends on the path difference $\delta = a + b - d$ (the path over the top minus the direct path) compared with the wavelength. In 1968 Maekawa measured the attenuation in the laboratory and found the simple relation

$$\Delta L \approx 10\log_{10}(3 + 20N), \qquad N = \frac{2\delta}{\lambda}$$

where N is the Fresnel number. Low frequencies with long wavelengths bend easily round the edge; high frequencies are attenuated a lot. In practice a barrier gives at most 15–20 dB, because sound is also scattered in the air above the barrier and reflected from buildings. The barrier must be airtight, weigh at least 10–20 kg/m², and preferably be absorbent on the road side.





Path difference δ = a + b − d = 61.0 cm

Fresnel number at 500 Hz: N = 1.78

Total attenuation of traffic noise: 18.1 dB(A)

Façade insulation

Where barriers cannot be used – in towns – the home must be protected at the façade. The window is almost always the weak link. An ordinary double-glazed window from the 1980s insulates about 25–30 dB; acoustic glazing with laminated glass of different thicknesses 35–40 dB. Ventilation openings and leaky joints can spoil the rest. The noise entering through a façade is calculated by adding the energies from the individual elements weighted by their areas (see building and room acoustics). Planning also helps: bedrooms on the quiet side and a continuous building towards the road that screens the courtyard behind.

Noise and health

In the 1980s traffic noise was mostly regarded as an annoyance. Today there is good evidence that long-term traffic noise increases the risk of sleep disturbance, high blood pressure, cardiovascular disease and diabetes – large Danish population studies from the Danish Cancer Society have contributed significantly. In 2018 the WHO recommended keeping road noise below $L_{\mathrm{den}} = 53$ dB and below 45 dB at night. The Danish EPA estimates that road noise contributes to several hundred premature deaths a year in Denmark.

Since 1989

- **Noise maps:** The EU Environmental Noise Directive (2002/49/EC) requires noise maps and action plans for major roads and cities every five years; the common calculation method CNOSSOS-EU has been used since 2019.

- **Low-noise asphalt:** Open and thin-layer surfaces, tested by the Danish Road Directorate since the 1990s, give 2–5 dB less tyre/road noise, but the effect declines as the pores clog.
- **Low-noise tyres:** Since 2012 tyres in the EU have carried a label stating their noise level.
- **Electric cars:** Below 20 km/h electric cars must emit a warning sound (AVAS) to protect pedestrians – here silence has become a problem.

Further reading

- Danish EPA: *Støj fra veje* (Road noise), guideline no. 4/2007 with later supplements (in Danish)
- WHO Regional Office for Europe: *Environmental Noise Guidelines for the European Region* (2018)
- Z. Maekawa: "Noise reduction by screens", *Applied Acoustics* 1 (1968) 157–173
- Jørgen Kragh et al.: *Nordic prediction method for road traffic noise* (1996) and Nord2000

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